An Extended Flux Model-Based Rotor Position Estimator for Sensorless Control of Salient-Pole Permanent-Magnet Synchronous Machines

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***Abstract*—Starting from the classical dynamic model of salient-pole permanent-magnet synchronous machines (PMSMs) expressed in the stationary reference frame, this paper presents a mathematical model reconstruction process for salient-pole PMSMs, from which an extended flux-based machine model is derived. Compared with the commonly used extended electromotive force-based model, the extended flux-based model has notable advantages of simpler model structure and less sensitive to the variations of machine parameters and operating conditions. A new extended flux model-based rotor position estimator is then proposed for sensorless control of salient-pole PMSMs by utilizing a sliding-mode observer with a dynamic position compensator. The latter improves the dynamic performance and low-speed operating capability of the sensorless control system. Both simulation and experimental results are provided to validate the proposed rotor position estimator and the sensorless control system for salient-pole PMSMs.**

***Index Terms*—Extended flux model, permanent-magnet synchronous machine (PMSM), position estimation, salient pole, sensorless control.**

I. Introduction

In recent years, much research effort has gone into the development of rotor position/speed sensorless drives that have comparable dynamic performance with respect to the sensor-based drives, i.e., drive systems with rotor position sensors, for permanent-magnet synchronous machines (PMSMs) [1]–[7], [10]–[12]. In the medium- and high-speed range, the model-based methods, e.g., the electromotive force (EMF)-based methods, are one of the most widely used strategies for rotor position estimation [1]–[3]. However, in the low-speed region, due to the small signal-to-noise ratio (SNR), i.e., the ratio between the magnitude of the EMF components and the magnitude of the noise, the EMF-based rotor position estimators are not accurate enough. Therefore, the capability of the EMF-based rotor position estimators should be further improved for low-speed operations.

Due to the machine rotor saliency, the rotor position estimation algorithm for a salient-pole PMSM is generally more complex than that for a nonsalient-pole PMSM, e.g., a surface-mounted PMSM. To perform the model-based rotor position estimation for salient-pole PMSMs, model reconstruction has been commonly performed to convert a salient-pole PMSM into an equivalent nonsalient-pole PMSM. Two categories of reconstructed salient-pole PMSM models, i.e., voltage- or EMF-based models and flux-based models, have been reported in the literature. The "extended EMF (EEMF)" model [1], [2] is the most widely used one for rotor position estimation. In the EEMF model, the saliency-related voltage terms are converted into EMF terms; the EEMF is then formed to be a summation of the saliency-related EMF terms and the original back EMF terms. In the EEMF model, only the EEMF components contain the rotor position information. However, since the amplitude of the EEMF components depends on operating conditions and changes of stator currents, the dynamic performance of an EEMF-based rotor position estimator may degrade during an abrupt change of the operating condition. Moreover, since the EEMF model needs the information of rotor speed and machine parameters, i.e., stator resistance and inductance, it is difficult to design an observer which is robust to both load condition variations and machine parameter uncertainties.

Besides the EEMF-based model, the flux-based models, e.g., the "fictitious flux" model [4] and the "active flux" model [5], provide alternatives to convert a salient-pole PMSM model into an equivalent nonsalient-pole PMSM model mathematically. Based on a flux model, a flux observer has been commonly designed to estimate the flux components [4], [5], from which the rotor position information can be extracted. In the flux model-based rotor position estimator, an integrator is normally required to calculate the flux terms. In this case, some practical issues, e.g., current sensor dc offset, integrator dc offset and initial condition, should be carefully handled. Although many EEMF model and flux-model-based rotor position estimators can be found in the literature, little work has been reported to compare these two types of models for rotor position estimation.

This paper presents a mathematical model reconstruction process for dynamic modeling of a generic salient-pole PMSM in

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the stationary reference frame. By reconstructing the machine model using the voltage concept, the EEMF-based model can be obtained; while by reconstructing the machine model using the flux concept, an extended flux-based salient-pole PMSM model can be derived. Compared to the EEMF model, the extended flux model has the advantages of simpler structure, independence to rotor speed, and less sensitivity to machine parameter uncertainties. Different from the active flux model in [5] derived in the rotor reference frame, the extended flux model is derived in the stationary reference frame, although the expression of the extended flux is equivalent to that of the active flux in [5].

Based on the extended flux model, a novel rotor position estimator is proposed by using a sliding-mode observer (SMO) with a dynamic position compensator for sensorless control of salient-pole PMSMs. Instead of extracting the rotor position information from the "active flux" estimated by a flux observer in [5], the SMO is designed to estimate the derivatives of the extended flux components, which are voltage components, from which the rotor position information is extracted. The dynamic position compensator is designed to account for the error of rotor position estimation due to the variations of the extended flux components to improve the transient performance and low-speed operating capability of the sensorless drive. Extensive simulation and experimental results on a 200-W salient-pole PMSM drive system are presented to validate the proposed rotor position estimator and sensorless control.

II. Model Reconstruction for Salient-Pole PMSMs

A. Dynamic Model of a Salient-Pole PMSM

The dynamics of a salient-pole PMSM can be modeled in the *dq* rotating reference frame as

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} R + pL_d & -\omega_{re} L_q \\ \omega_{re} L_d & R + pL_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_{re}\lambda_m \end{bmatrix} \tag{1}$$

where p is a derivative operator; v_d and v_q are the *d*-axis and *q*-axis stator voltages, respectively; i_d and i_q are the *d*-axis and *q*-axis stator currents, respectively; ω_{re} is the rotor electrical speed; L_d and L_q are the *d*-axis and *q*-axis inductances, respectively; λ_m is the flux linkage produced by the permanent magnets, and R is the stator resistance. Using the inverse Park transformation, the salient-pole PMSM model in the $\alpha\beta$ stationary reference frame can be expressed as

$$\begin{aligned} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = {} & R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \\ & + \begin{bmatrix} L + \Delta L\cos(2\theta_{re}) & \Delta L \sin(2\theta_{re}) \\ \Delta L\sin(2\theta_{re}) & L - \Delta L\cos(2\theta_{re}) \end{bmatrix} \\ & \cdot p \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \omega_{re}\lambda_m \begin{bmatrix} -\sin\theta_{re} \\ \cos\theta_{re} \end{bmatrix} \end{aligned} \tag{2}$$

where $L = (L_d + L_q)/2, \Delta L = (L_d - L_q)/2$, and θ_{re} is the rotor position angle. Due to the difference between L_d and L_q caused by machine rotor saliency, both θ_{re} and $2\theta_{re}$ terms appear in (2). Therefore, it is difficult to use (2) directly for rotor position observation. A reconstruction of (2) is needed to facilitate the rotor position observation for a salient-pole PMSM.

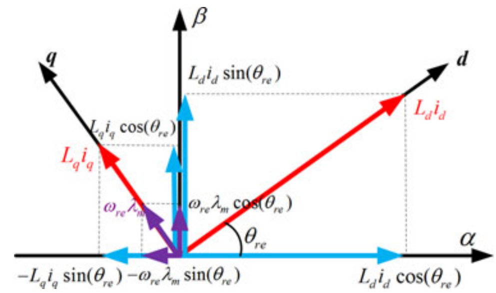

Fig. 1. Illustration of the salient-pole PMSM model (3).

This paper proposes to reconstruct the salient-pole PMSM model mathematically from a voltage/flux model as follows:

$$\begin{aligned} v_\alpha &= Ri_\alpha \\ &+ \overbrace{L_d p\left(i_d\cos(\theta_{re})\right) - L_q p\left(i_q \sin(\theta_{re})\right) + \lambda_m p\left(\cos(\theta_{re})\right)}^{p(\lambda_\alpha)} \\ v_\beta &= Ri_\beta \\ &+ \underbrace{L_d p\left(i_d\sin(\theta_{re})\right) + L_q p\left(i_q \cos(\theta_{re})\right) + \lambda_m p\left(\sin(\theta_{re})\right)}_{p(\lambda_\beta)}. \end{aligned} \tag{3}$$

Equation (3), which models the voltage/flux dynamics of the PMSM in the stationary reference frame, contains the voltage terms (v_α and v_β) in the stationary reference frame and the derivatives of the flux terms ($p\lambda_\alpha$ and $p\lambda_\beta$) expressed with quantities in the *dq* rotating reference frame. In (3), only the θ_{re} related terms are present, and each term has clear physical meaning, as shown in Fig. 1. Rearranging (3), the following equations can be obtained:

$$\begin{aligned} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} &= R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + p \begin{bmatrix} L & 0 \\ 0 & L \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \\ & \underbrace{+ \Delta L p\left(\begin{bmatrix} \cos\theta_{re} & \sin\theta_{re} \\ \sin\theta_{re} & -\cos\theta_{re} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \right) + \omega_{re}\lambda_m \begin{bmatrix} -\sin\theta_{re} \\ \cos\theta_{re} \end{bmatrix}}_{\text{Position Related Terms } V(\theta_{re})}. \end{aligned} \tag{4}$$

B. The Idea of Model Reconstruction

To facilitate the rotor position observation, the objective of reconstructing the model (3) is to achieve a similar symmetrical model structure, which contains a symmetrical inductance matrix as for the nonsalient-pole PMSMs as follows:

$$\begin{aligned} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} &= R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + p \begin{bmatrix} L_s & 0 \\ 0 & L_s \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \\ &\quad + \omega_{re}\lambda_m \begin{bmatrix} -\sin\theta_{re} \\ \cos\theta_{re} \end{bmatrix} \\ &= R \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + p \begin{bmatrix} L_s & 0 \\ 0 & L_s \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + p \begin{bmatrix} \lambda_m\cos\theta_{re} \\ \lambda_m\sin\theta_{re} \end{bmatrix}. \end{aligned} \tag{5}$$

In (5), the $\sin(\theta_{re})$ and $\cos(\theta_{re})$ related terms are present separately in each equation. However, in (4), both the $\sin(\theta_{re})$ and $\cos(\theta_{re})$ related terms are present simultaneously in each equation. Therefore, further model reconstruction is required for (4) to achieve a similar model structure as (5). As shown in (5), the EMF term can be either written in the form of voltage, i.e., $\omega_{re}\lambda_m[-\sin(\theta_{re}), \cos(\theta_{re})]^T$, or in the form of derivative of flux, i.e., $p[\lambda_m \cos(\theta_{re}), \lambda_m \sin(\theta_{re})]^T$. Similarly, (4) can be further reconstructed in either a voltage (EMF) form or a flux form.

C. Model Reconstruction Based on Voltage Concept

Consider the last two terms of (4) as follows, which are position related:

$$V(\theta_{re}) = \begin{bmatrix} \Delta Lp(i_d \cos\theta_{re} + i_q \sin\theta_{re}) - \omega_{re}\lambda_m \sin\theta_{re} \\ \Delta Lp(i_d \sin\theta_{re} - i_q \cos\theta_{re}) + \omega_{re}\lambda_m \cos\theta_{re} \end{bmatrix}. \tag{6-1}$$

By applying the following inverse Park transformation to the currents

$$\begin{cases} i_\alpha = i_d \cos\theta_{re} - i_q \sin\theta_{re} \\ i_\beta = i_d \sin\theta_{re} + i_q \cos\theta_{re} \end{cases} \tag{6-2}$$

(6-1) can be reconstructed into the following form:

$$V(\theta_{re}) = \begin{bmatrix} \Delta Lp(i_\alpha + 2i_q \sin\theta_{re}) - \omega_{re}\lambda_m \sin\theta_{re} \\ \Delta Lp(i_\beta - 2i_q \cos\theta_{re}) + \omega_{re}\lambda_m \cos\theta_{re} \end{bmatrix}. \tag{6-3}$$

In (6-3), the $\sin(\theta_{re})$ and $\cos(\theta_{re})$ related terms are present separately in each equation. However, both voltage terms, e.g., $\omega_{re}\lambda_m \sin(\theta_{re})$, and derivative of flux terms, e.g., $p[\Delta L i_q \sin(\theta_{re})]$, are still present in each equation. Since $\sin(\theta_{re})$ and its derivative cannot be combined directly [neither $\cos(\theta_{re})$ and its derivative], the derivative of flux terms, e.g., $p[\Delta L i_q \sin(\theta_{re})]$, need to be converted into voltage terms in order to complete the reconstruction of (6-3) into the voltage form.

Applying (6-2) two more times to (6-3), the following equations can be obtained as (6-4), shown at the bottom of the page.

Equations (6-4) are a part of the EEMF model proposed in [1].

D. Model Reconstruction Based on Flux Concept

The last two terms of (4) can be reconstructed as follows:

$$\begin{aligned} V(\theta_{re}) = {} & \Delta Lp\left(\begin{bmatrix} \cos\theta_{re} & \sin\theta_{re} \\ \sin\theta_{re} & -\cos\theta_{re} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix}\right) \\ & + p\begin{bmatrix} \lambda_m \cos\theta_{re} \\ \lambda_m \sin\theta_{re} \end{bmatrix} \\ = {} & \begin{bmatrix} \Delta Lp(i_d \cos\theta_{re} + i_q \sin\theta_{re}) + p(\lambda_m \cos\theta_{re}) \\ \Delta Lp(i_d \sin\theta_{re} - i_q \cos\theta_{re}) + p(\lambda_m \sin\theta_{re}) \end{bmatrix}. \end{aligned} \tag{7-1}$$

By using (6-2), (7-1) can be reconstructed into the following form:

$$V(\theta_{re}) = \begin{bmatrix} \Delta Lp(2i_d \cos\theta_{re} - i_\alpha) + p(\lambda_m \cos\theta_{re}) \\ \Delta Lp(2i_d \sin\theta_{re} - i_\beta) + p(\lambda_m \sin\theta_{re}) \end{bmatrix}. \tag{7-2}$$

Different from (6-3), only the derivative of flux terms, e.g., $p[\lambda_m \cos(\theta_{re})]$, are present in (7-2). Rearranging (7-2) yields

$$\begin{aligned} V(\theta_{re}) = {} & \begin{bmatrix} -\Delta Lp(i_\alpha) + p\left[(\lambda_m + 2\Delta L i_d)\cos\theta_{re}\right] \\ -\Delta Lp(i_\beta) + p\left[(\lambda_m + 2\Delta L i_d)\sin\theta_{re}\right] \end{bmatrix} \\ = {} & \begin{bmatrix} -\Delta Lp(i_\alpha) + p\left[\lambda_{\text{ext}} \cos\theta_{re}\right] \\ -\Delta Lp(i_\beta) + p\left[\lambda_{\text{ext}} \sin\theta_{re}\right] \end{bmatrix} \end{aligned} \tag{7-3}$$

where $\lambda_{\text{ext},\alpha\beta} = \lambda_{\text{ext}} \cdot [\cos\theta_{re}, \ \sin\theta_{re}]^T$ is the position-related flux vector, which is defined as the extended flux, and $\lambda_{\text{ext}} = \lambda_m + 2\Delta L i_d = \lambda_m + (L_d - L_q)i_d$ is the magnitude of the extended flux.

E. Reconstructed Salient-Pole PMSM Models

Substituting (6-4) and (7-3) into (4) yields the following EEMF model (8) proposed in [1] and the extended flux model (9), respectively:

$$\begin{aligned} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = {} & \begin{bmatrix} R & \omega_{re}(L_d - L_q) \\ -\omega_{re}(L_d - L_q) & R \end{bmatrix} \cdot \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \\ & + p\begin{bmatrix} L_d & 0 \\ 0 & L_d \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + E_{\text{ext}} \begin{bmatrix} -\sin\theta_{re} \\ \cos\theta_{re} \end{bmatrix} \end{aligned} \tag{8}$$

where $E_{\text{ext}} = \omega_{re}[\lambda_m + (L_d - L_q)i_d] - (L_d - L_q)p(i_q)$ is the amplitude of the position-related EEMF components.

$$\begin{aligned} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = {} & \begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix} \cdot \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \\ & + p\begin{bmatrix} L_q & 0 \\ 0 & L_q \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + p\begin{bmatrix} \lambda_{\text{ext}} \cos\theta_{re} \\ \lambda_{\text{ext}} \sin\theta_{re} \end{bmatrix}. \end{aligned} \tag{9}$$

A comparison among the nonsalient-pole PMSM model (5) and the two salient-pole PMSM models, i.e., the EEMF model (8) and extended flux model (9), is provided in Table I. Moreover, the following is the "active flux" model presented in [5],

$$\begin{aligned} V(\theta_{re}) = {} & \begin{bmatrix} \Delta Lp(i_\alpha) + 2\omega_{re}\Delta L i_q \cos\theta_{re} - (\omega_{re}\lambda_m - 2\Delta Lp(i_q))\sin\theta_{re} \\ \Delta Lp(i_\beta) + 2\omega_{re}\Delta L i_q \sin\theta_{re} + (\omega_{re}\lambda_m - 2\Delta Lp(i_q))\cos\theta_{re} \end{bmatrix} \\ = {} & \begin{bmatrix} \Delta Lp(i_\alpha) + 2\omega_{re}\Delta L i_\beta - [\omega_{re}(\lambda_m + 2\Delta L i_d) - 2\Delta Lp(i_q)]\sin\theta_{re} \\ \Delta Lp(i_\beta) - 2\omega_{re}\Delta L i_\alpha + [\omega_{re}(\lambda_m + 2\Delta L i_d) - 2\Delta Lp(i_q)]\cos\theta_{re} \end{bmatrix} \end{aligned} \tag{6-4}$$

TABLE I
COMPARISON AMONG (5), (8), AND (9)

Machine type	Machine model based on voltage concept			Machine model based on flux concept		
	Impedance matrix	Inductance matrix	Position-related terms	Impedance matrix	Inductance matrix	Position-related terms
Nonsalient-pole PMSM	$\begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix}$	$\begin{bmatrix} L_s & 0 \\ 0 & L_s \end{bmatrix}$	$\omega_{re}\lambda_m \begin{bmatrix} -\sin\theta_{re} \\ \cos\theta_{re} \end{bmatrix}$	$\begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix}$	$\begin{bmatrix} L_s & 0 \\ 0 & L_s \end{bmatrix}$	$p\begin{bmatrix} \lambda_m \cos\theta_{re} \\ \lambda_m \sin\theta_{re} \end{bmatrix}$
Salient-pole PMSM	$\begin{bmatrix} R & 2\Delta L\omega_{re} \\ -2\Delta L\omega_{re} & R \end{bmatrix}$	$\begin{bmatrix} L_d & 0 \\ 0 & L_d \end{bmatrix}$	$E_{\text{ext}} \begin{bmatrix} -\sin\theta_{re} \\ \cos\theta_{re} \end{bmatrix}$	$\begin{bmatrix} R & 0 \\ 0 & R \end{bmatrix}$	$\begin{bmatrix} L_q & 0 \\ 0 & L_q \end{bmatrix}$	$p\begin{bmatrix} \lambda_{\text{ext}} \cos\theta_{re} \\ \lambda_{\text{ext}} \sin\theta_{re} \end{bmatrix}$

which was derived and expressed in the rotor reference frame:

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} R_s & -\omega_{re}L_q \\ \omega_{re}L_q & R_s \end{bmatrix}\begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} L_q & 0 \\ 0 & L_q \end{bmatrix} \cdot p\begin{bmatrix} i_d \\ i_d \end{bmatrix} + \begin{bmatrix} p\lambda_{\text{ext}} \\ \omega_{re}\lambda_{\text{ext}} \end{bmatrix}. \tag{10}$$

Although the definitions of the extended flux in (9) and the active flux in (10) are equivalent, the derivations of these two models are originated from difference concepts and performed in different reference frames.

In addition, a comparison between (8) and (9) from the observer design aspect is provided as follows:

1) A rotor position observer based on (8) needs the values of all machine parameters, including R, L_d, and L_q. However, a rotor position observer based on (9) does not need L_d information.
2) In (8), both v_α and v_β are functions of i_α and i_β. Therefore, the α- and β-loops are not completely decoupled. However, in (9), v_α is a function of i_α only, and v_β is a function of i_β only. Therefore, the α- and β-loops are decoupled.
3) In (8), the speed information, ω_{re}, is needed; while (9) does not need ω_{re}.
4) The E_{ext} in (8) depends on both ω_{re} and $p(i_q)$ and, thus, is sensitive to load variations. This may degrade the dynamic performance of the observer. On the contrary, λ_{ext} in (9) depends on neither ω_{re} nor $p(i_q)$. Therefore, an observer based on (9) should have better dynamic performance.
5) An observer can be designed based on (8) to obtain the EEMF components, from which the rotor position can be estimated directly. However, an observer based on (9) can only estimate the derivatives of the extended flux, and integration is needed to calculate the extended flux components, from which the rotor position can be obtained.

In summary, an observer based on (9) is less sensitive to machine parameters, speed, and load variations than that based on (8). However, an integrator is required to work with the observer together to calculate the extended flux components, from which the rotor position can be extracted directly.

III. Sliding-Mode Observer With Dynamic Position Compensator for Rotor Position Estimation

To design an observer based on (9) without using an integrator, the most straightforward approach is to further process the derivative of the extended flux to obtain a voltage term that contains the rotor position explicitly. The derivative of the extended flux can be viewed as a voltage term or EMF term, which is denoted as $e'_{\alpha\beta}$ and can be calculated as

$$\begin{aligned} e'_{\alpha\beta} = \begin{bmatrix} e'_\alpha \\ e'_\beta \end{bmatrix} &= p\begin{bmatrix} \lambda_{\text{ext}}\cos\theta_{re} \\ \lambda_{\text{ext}}\sin\theta_{re} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{re}p\left(\lambda_{\text{ext}}\right) + \lambda_{\text{ext}}p\left(\cos\theta_{re}\right) \\ \sin\theta_{re}p\left(\lambda_{\text{ext}}\right) + \lambda_{\text{ext}}p\left(\sin\theta_{re}\right) \end{bmatrix} \\ &= \omega_{re}\lambda_{\text{ext}}\begin{bmatrix} -\sin\theta_{re} \\ \cos\theta_{re} \end{bmatrix} \\ &\quad + \left(L_d - L_q\right)p(i_d)\begin{bmatrix} \cos\theta_{re} \\ \sin\theta_{re} \end{bmatrix}. \end{aligned} \tag{11}$$

As shown in (11), both $\cos\theta_{re}$ and $\sin\theta_{re}$ related terms are present simultaneously in the expressions of e'_α and e'_β. Therefore, it is still complex to estimate the rotor position using (11) directly. However, as presented in [6], if the variation of i_d is ignored, the last term in (11) can be ignored such that the rotor position estimation will be notably simplified. However, the method in [6] has obvious limitation due to the assumption that $p(i_d)$ is negligible or $(L_d - L_q)\cdot p(i_d) \ll \omega_{re}\lambda_{\text{ext}}$. To eliminate the limitation, $e'_{a\beta}$ is processed in the phasor form as follows:

$$\begin{aligned} &e'_\beta - je'_\alpha \\ &= \omega_{re}\lambda_{\text{ext}}(\cos\theta_{re} + j\sin\theta_{re}) \\ &\quad + p(\lambda_{\text{ext}})(-j\cos\theta_{re} + \sin\theta_{re}) \\ &= \omega_{re}\lambda_{\text{ext}}e^{j\theta_{re}} - jp\left(\lambda_{\text{ext}}\right)e^{j\theta_{re}} = Ae^{j(\theta_{re}+\varphi)} \end{aligned} \tag{12}$$

where $A = \sqrt{(p(\lambda_{\text{ext}}))^2 + (\omega_{re}\lambda_{\text{ext}})^2}$ is the transient amplitude and $\varphi = \tan^{-1}\left[\frac{-p(\lambda_{\text{ext}})}{\omega_{re}\lambda_{\text{ext}}}\right]$ is the disturbance phase angle. If $p(\lambda_{\text{ext}}) = 0$, $|\varphi|$ will be equal to zero, which means that the rotor position calculated from $e'_{\alpha\beta}$, $\tilde{\theta} = \tan^{-1}\left[-\frac{p(\lambda_{\text{ext}}\cos\theta_{re})}{p(\lambda_{\text{ext}}\sin\theta_{re})}\right]$, is equal to the actual rotor position θ_{re}. However, in practical

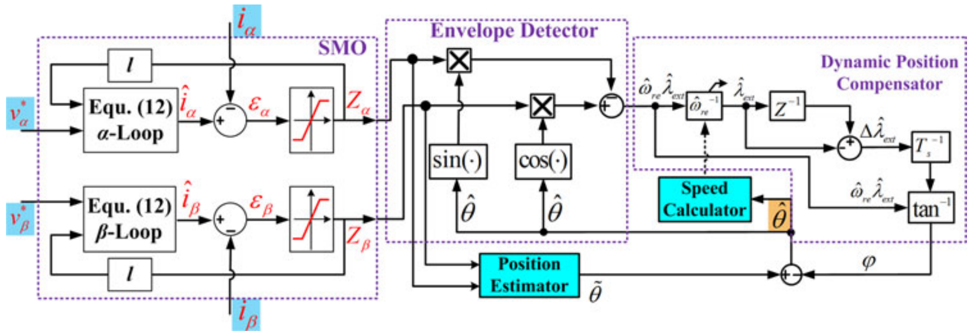

Fig. 2. Schematic diagram of the proposed rotor position estimator.

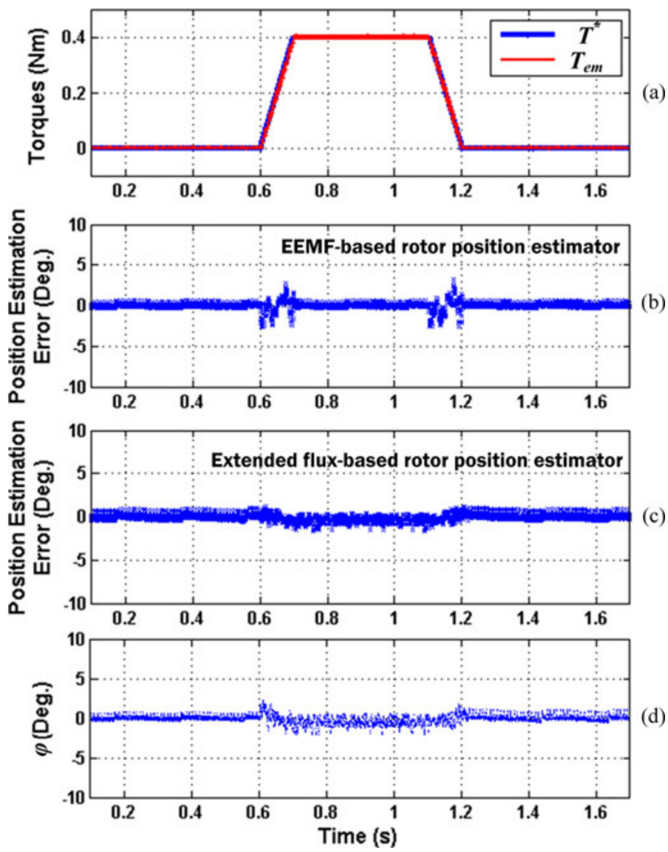

Fig. 3. Comparison of the proposed extended flux-based rotor position estimator and the EEMF-based rotor position estimator when the salient-pole PMSM operates at the base speed with low slew rate torque variations.

applications when the salient-pole PMSM operates in the low-speed region or has a large variation of the extended flux (e.g., caused by an abrupt change of i_d), φ will not be exactly equal to zero and a phase error will exist.

This paper proposes to design an SMO based on the extended flux model (9) to estimate the extended flux components, from which $\tilde{\theta}$ can be obtained. Since $\tilde{\theta}$ is not an accurate estimation of the actual rotor position, a dynamic position compensator is further proposed to eliminate the error between $\tilde{\theta}$ and the actual rotor position to improve the rotor position estimation performance during low-speed operations and large load transients.

The overall block diagram of the proposed rotor position estimator is shown in Fig. 2, which contains three major parts: an SMO, an envelope detector, and a dynamic position compensator. To utilize digital controllers for salient-pole PMSM drives, a discrete-time SMO [13] is designed according to (9) as follows:

$$\begin{cases} \hat{i}_\alpha[k+1] = T_s\left(\dfrac{v_\alpha^*}{L_q} + lZ_\alpha[k]\right) + \left(1 - \dfrac{T_s R}{L_q}\right)\hat{i}_\alpha[k] \\ \hat{i}_\beta[k+1] = T_s\left(\dfrac{v_\beta^*}{L_q} + lZ_\beta[k]\right) + \left(1 - \dfrac{T_s R}{L_q}\right)\hat{i}_\beta[k] \end{cases} \tag{13}$$

where T_s is the sampling period of the SMO; v_α^* and v_β^* are the voltage commands generated by the current controllers [8], [9]; $Z_\alpha[k]$ and $Z_\beta[k]$ are the outputs of the switching function at the kth time step, which contain $e'_{\alpha\beta}$ components, if the sliding mode is enforced. The angle between the vector $e'_\beta - je'_\alpha$ and the α-axis can be estimated as $\tilde{\theta} = \tan^{-1}\left(-\frac{Z_\alpha}{Z_\beta}\right)$.

However, per previous discussion, $\tilde{\theta}$ needs to be compensated for the disturbance phase error φ in the low-speed and transient conditions, and φ can be calculated as follows:

$$\varphi[k] = \tan^{-1}\left[\frac{\hat{\lambda}_{\text{ext}}[k-1] - \hat{\lambda}_{\text{ext}}[k]}{\hat{\omega}_{re}[k]\hat{\lambda}_{\text{ext}}[k]T_s}\right] \tag{14}$$

where $\hat{\omega}_{re}$ and $\hat{\lambda}_{\text{ext}}$ are the estimated rotor speed and magnitude of the estimated extended flux, respectively. A dynamic position compensator, as shown in Fig. 2, is designed to obtain φ based on (14). The estimated rotor position $\hat{\theta}$ is obtained by adding φ to $\tilde{\theta}$. The estimated rotor speed $\hat{\omega}_{re}$ can then be obtained from $\hat{\theta}$ by using a moving average or phase-locked loop (PLL) method [14].

An envelope detector is designed to estimate the product of $\hat{\omega}_{re}$ and $\hat{\lambda}_{\text{ext}}$ in (14), which can be expressed as

$$\begin{aligned} &Z_\beta \cos\hat{\theta} - Z_\alpha \sin\hat{\theta} \\ &\approx \hat{\omega}_{re}\hat{\lambda}_{\text{ext}} + p\left(\lambda_{\text{ext}}\right)\sin\left(\theta_{re} - \hat{\theta}\right) \xrightarrow{\theta_{re}\approx\hat{\theta}} \approx \hat{\omega}_{re}\hat{\lambda}_{\text{ext}}. \end{aligned} \tag{15}$$

According to (15), if the error between the estimated and actual rotor positions is small enough, the $\sin(\theta_{re} - \hat{\theta})$ term can be ignored, such that $\hat{\omega}_{re}\hat{\lambda}_{\text{ext}}$ is obtained. Once the rotor speed is estimated, the value of $\hat{\lambda}_{\text{ext}}$ can then be calculated.

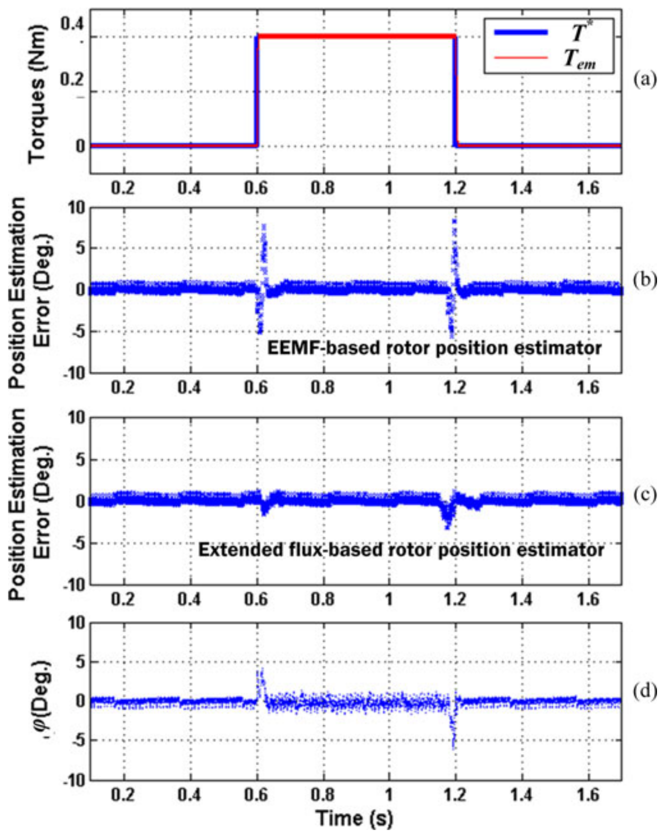

Fig. 4. Comparison of the proposed extended flux-based rotor position estimator and the EEMF-based rotor position estimator when the salient-pole PMSM operates at the base speed with high slew rate torque variations.

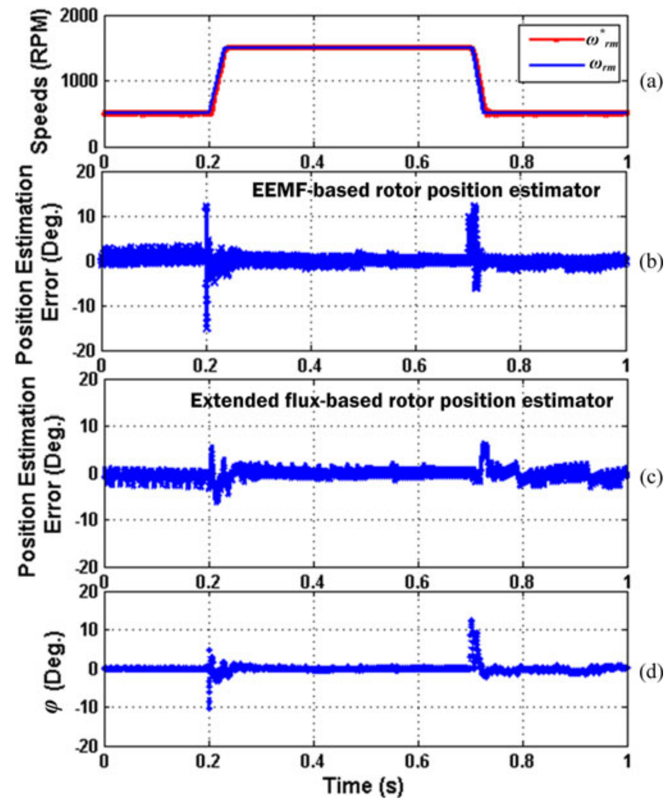

Fig. 5. Speed control performance of the sensorless PMSM drive. (a) Commanded and measured rotor speeds of the sensorless drive with the proposed rotor position estimator (rotor speed changes between 500 and 1500 r/min), (b) rotor position estimation error of the EEMF-based estimator, (c) rotor position estimation error of the proposed estimator, and (d) disturbance phase angle φ obtained from the dynamic position compensator.

IV. Simulation Results

Per previous discussion, the proposed rotor position estimator and sensorless drive system should have better dynamic performance and low-speed operating capability than the EEMF-based methods. In this section, simulation studies are performed to compare the performance of the proposed rotor position estimator and the EEMF-based rotor position estimator proposed in [13]. The parameters of the salient-pole PMSM used in the simulation are: $R = 0.23\ \Omega$, $L_d = 0.275$ mH, $L_q = 0.364$ mH, $\lambda_m = 0.0095$ Vs/rad, rated power $=$ 200 W, base speed $=$ 1500 r/min, maximum speed $=$ 3000 r/min, maximum amplitude of stator currents 5 A, and the number of pole pairs $= 4$. In this section, typical simulation results when the PMSM operates in torque control mode and speed control mode will be presented. In addition, the low-speed operating capability of the proposed sensorless drive system will also be investigated.

Fig. 3 compares the performance of the two rotor position estimators when the salient-pole PMSM operates at 1500 r/min with low slew rate torque changes. The commanded torque (T^*) and generated electromagnetic torque (T_{em}) of the PMSM using the proposed rotor position estimator are shown in Fig. 3(a). The generated electromagnetic torque of the PMSM can well track the torque command during the whole test. The position estimation errors of the EEMF-based rotor position estimator and proposed rotor position estimator are shown in Fig. 3(b) and (c), respectively. When a low slew rate torque change is applied, the variation of the magnitude of the extended flux is quite small. Consequently, the disturbance phase angle φ obtained from the dynamic position compensator is almost zero and negligible, as shown in Fig. 3(d); and the proposed rotor position estimator is only slightly better than the EEMF-based rotor position estimator during torque transients. However, when the torque has high slew rate changes, e.g., step changes shown in Fig. 4(a), the extended flux has large variations. Consequently, the disturbance phase angle φ obtained from the dynamic position compensator is large during the torque transients, as shown in Fig. 4(d). With proper phase compensation using φ, the rotor position estimation error of the proposed estimator is significantly smaller than that of the EEMF-based estimator during the torque transients, as compared in Fig. 4(b) and (c).

Fig. 5 compares the speed tracking performance of sensorless PMSM drive with two different rotor position estimators when the PMSM is operated in the speed control mode at the no load condition. The rotor speed response of the sensorless drive equipped with the proposed rotor position estimator is shown in Fig. 5(a), where ω^*_{rm} and ω_{rm} are the commanded and measured rotor mechanical speeds, respectively. The rotor speed firstly increases from 500 r/min to the base speed of 1500 r/min within 20 ms and stays at the base speed; then the rotor speed ramps

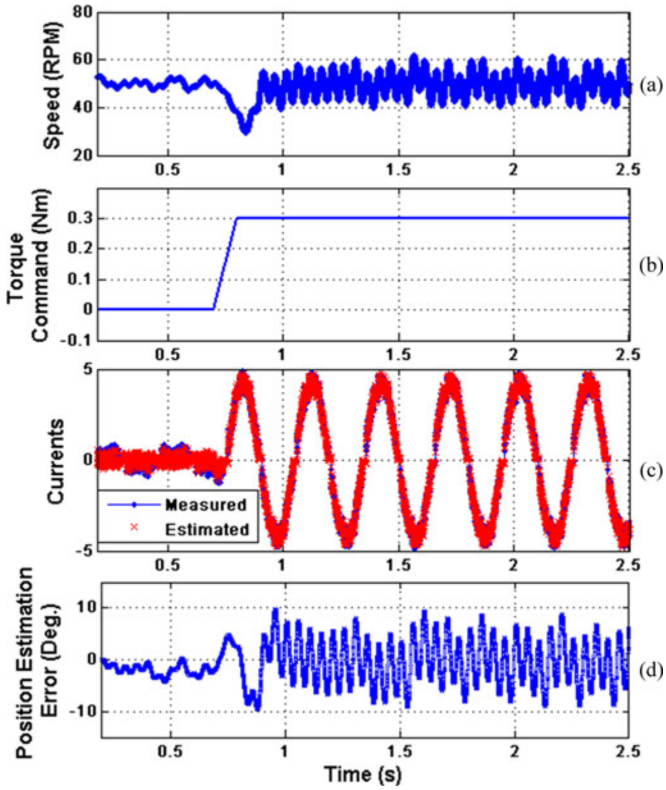

Fig. 6. Torque control performance of the sensorless PMSM drive with the proposed rotor position estimator at 50 r/min (1.67% of the maximum speed). (a) Measured rotor speed, (b) torque command, (c) estimated and measured α-axis stator currents, and (d) error between the estimated and measured rotor positions.

down from 1500 to 500 r/min within 20 ms. The corresponding position estimation errors of the EEMF-based rotor position estimator and the proposed rotor position estimator are shown in Fig. 5(b) and (c), respectively. The profile of φ is shown in Fig. 5(d). The results show that during the speed transition, with the help of the dynamic phase compensator, the amplitude of the rotor position estimation error of the proposed estimator is much smaller than that of the EEMF-based estimator.

To verify the low-speed operating capability of the sensorless drive using the proposed estimator, a torque ramp up test is performed at 50 r/min, which is 1.67% of the maximum speed. The corresponding simulation results are presented in Fig. 6. The EEMF-based rotor position estimator failed in this case. However, the proposed rotor position estimator still works and the accuracy of the rotor position estimation is still acceptable, as shown in Fig. 6(d).

V. Experimental Results

A. Test Stand Setup

An experimental test stand, as shown in Fig. 7(a), is designed to validate the proposed rotor position/speed sensorless control system. The schematic of the overall test stand setup is shown in Fig. 7(b), which includes a 200-W salient-pole PMSM connected back to back with a 200-W dc machine. The PMSM is

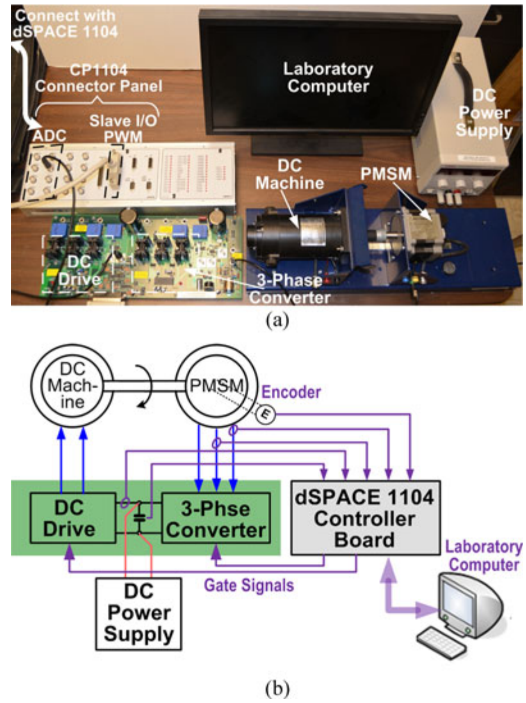

Fig. 7. Test stand. (a) The setup and (b) the schematic.

TABLE II
Specifications of the DC Motor and the Test Salient-Pole PMSM

Specifications	DC machine	Salient-pole PMSM
Maximum speed	3500 r/min	3000 r/min
Rated power	200 W	200 W
EMF constant	0.0087 V/r/min	0.0095 V/r/min
Stator resistance	0.39 Ω	0.23 Ω
Inductance(s)	Armature inductance 0.67 mH	$L_d = 0.275$ mH $L_q = 0.364$ mH
# of pole pairs	N/A	4

the same as that used in simulation studies. The dc machine can work as either a prime mover machine (motor) or a load machine (dc generator). The two machines share one common dc bus, whose voltage is maintained at 40 V by a dc power supply. The specifications of the dc motor and the PMSM are listed in Table II. The overall control algorithm is implemented in a dSPACE 1104 real-time control system. The PWM switching frequency at the base speed is 5 kHz. The phase currents are sampled twice per PWM cycle and the main control software (e.g., basic vector control, rotor position estimation, etc.) is also executed twice per PWM cycle. On the test stand, two phase stator currents of the PMSM and the dc bus current and voltage are measured by using current and voltage transducers, respectively. Four analog to digital converter (ADC) ports on the connector panel CP1104 are used for current and voltage sensing. The measured phase currents are used for current control and rotor position estimation of the PMSM. The PWM gate signals generated by the control system are sent to the converter drive circuit board through the slave I/O PWM connector on the CP1104. All of the experimental results are recorded by using

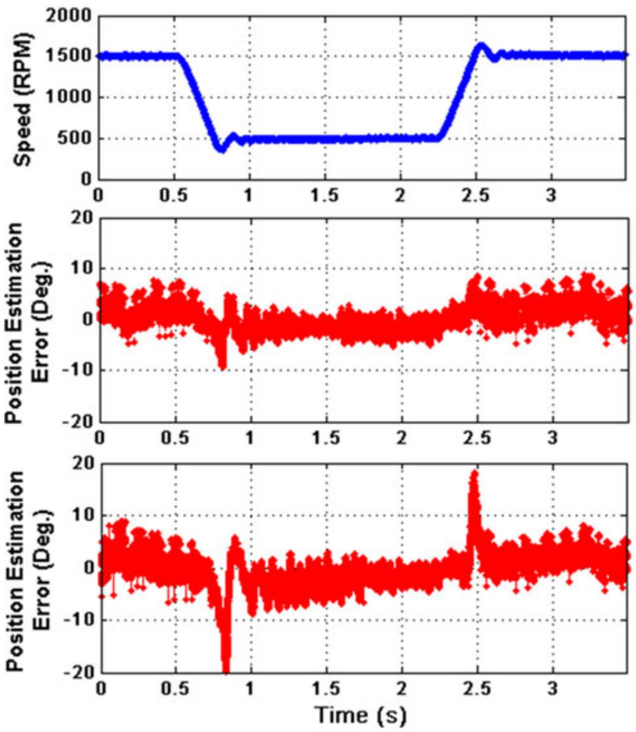

Fig. 8. Results for speed ramp change test. (a) Rotor speed profile, and error between the measured rotor position and the rotor position obtained from (b) the proposed estimator and (c) the EEMF-based estimator.

the ControlDesk installed on a laboratory computer, which is connected with the dSPACE system.

Two different rotor position estimators, i.e., the proposed rotor position estimator and the EEMF-based rotor position estimator proposed in [13], are both implemented in the control software. In the experiments, when the rotor position estimated by one of the two position estimators is used as the control angle, the other position estimator is disabled. The rotor position is also measured by an encoder, which is mounted on the rotor shaft of the PMSM. However, the measured rotor position is only used for evaluation purpose and is not used by the control algorithm.

B. Experimental Results

The performances of the two rotor position estimators are compared under the same speed ramp change test, where the salient-pole PMSM is operated in the speed control mode as a motor. The rotor speed first decreases from the rated value of 1500 to 500 r/min within 200 ms and then returns to the rated value within 200 ms. During the test, the dc machine is operated as a load machine to control the load torque of the PMSM to be around 70% of the maximum torque. The corresponding experimental results are shown in Fig. 8, In the steady states, when the speed command is fixed at either 1500 or 500 r/min, the rotor position estimation performances of the two estimators are similar. However, during the speed transient, the error between the measured rotor position and the rotor position obtained from the proposed estimator is much smaller than that obtained from the EEMF-based position estimator. These results verified that the transient performance of the proposed estimator is better than that of the EEMF-based position estimator.

To verify the low-speed operating capability of the sensorless drive using the proposed estimator, the experimental results at different rotor speeds below the base speed, including 1050, 300, and 100 r/min, are presented in Fig. 9. In these tests, the salient-pole PMSM is operated in the torque control mode with the maximum output torque applied; and the rotor speed of the PMSM is maintained to be constant by the prime mover. In each case, stable control is achieved and the estimated α-axis stator current well tracks its measured value.

To further validate the low-speed operating capability at an even lower speed point, the system is tested at 20% load torque and 50 r/min condition, which is only 1.67% of the maximum speed. The corresponding experimental results are presented in Fig. 10. The rotor speed has relatively larger ripples compared to those at 500 and 1500 r/min conditions. However, the average value of the rotor speed is maintained at 50 r/min. The estimated value of the α-axis PMSM stator current (i_α) well tracks the measured value, as shown in Fig. 10(b). The rotor position estimation error is limited within an acceptable range such that a stable speed control is maintained. While using the EEMF-based estimator, the speed control failed in such a low-speed condition due to the low SNR. In addition, the experimental result at 50 r/min with 100% load torque is shown in Fig. 11. Again, the PMSM stator current i_α well tracks the measured value and the rotor position estimation performance is similar to that in Fig. 10 in this heavy torque condition. Fig. 12 shows the steady-state results of the sensorless drive using the proposed position estimator without the dynamic position compensator under the same operating condition as in Fig. 10. As shown in Fig. 12, the amplitude of the position estimation error is larger than that shown in Fig. 10(c). This result validated the improvement of the sensorless drive in the low-speed operation by the proposed dynamic position compensator.

The experimental results of the sensorless torque control of the PMSM using the proposed rotor position estimator are presented in Figs. 13 and 14. These results are provided to evaluate the torque control performance of the proposed sensorless control system, and also to further validate the effectiveness of the dynamic phase compensator. In the test, the dc motor is operated as a prime mover machine, which regulates the shaft speed of the system. The salient-pole PMSM works as a generator in the torque control mode. The current commands are generated in different ways for the tests in Figs. 13 and 14. In Fig. 13, the i_d command is always set as zero, while i_q command is determined from the torque command. Since i_d command is a constant, the effect of the i_d variation on the variation of the extended flux can be ignored. While in Fig. 14, the i_d and i_q commands are generated by following the maximum torque per ampere (MTPA) profile. Since the i_d command is no longer a constant during the torque change, the amplitude of φ during the torque transient shall be much larger than that in Fig. 13.

There are three key points in the test shown in Fig. 14: (1) the dSPACE system switches from the edit mode to the animation

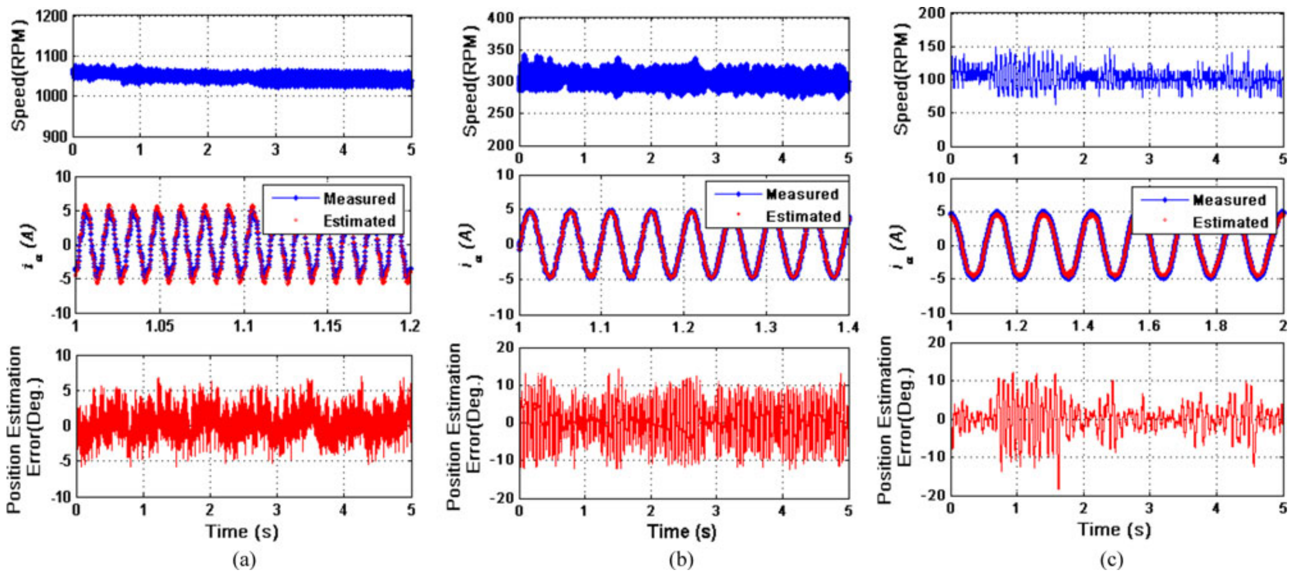

Fig. 9. Experimental results of sensorless control system at different speed points. (a) 1050 r/min, (b) 300 r/min, and (c) 100 r/min.

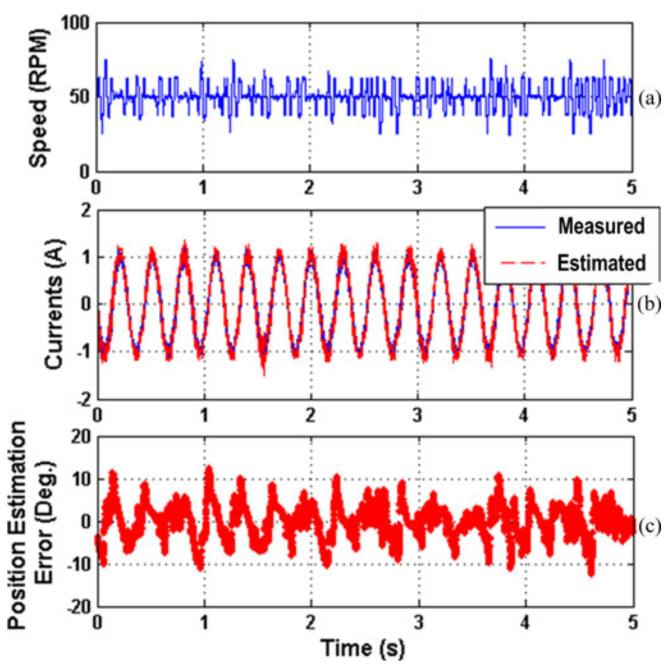

Fig. 10. Experimental results at 50 r/min with 20% load torque using the proposed rotor position estimator with the dynamic position compensator. (a) Rotor speed, (b) estimated and measured values of the α-axis stator current, and (c) error between the estimated and measured rotor positions.

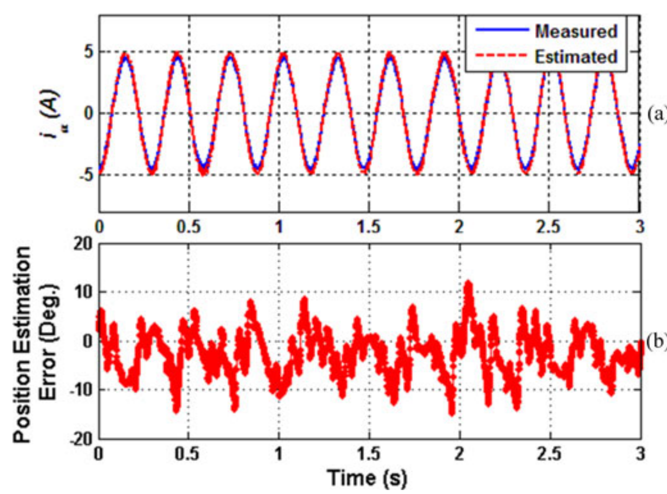

Fig. 11. Experimental results at 50 r/min with the maximum load torque using the proposed rotor position estimator with the dynamic position compensator. (a) Estimated and measured values of the α-axis stator current, and (b) error between the estimated and measured rotor positions.

mode; (2) the dc drive is enabled; and (3) the period of the PMSM torque changes at a constant speed. Before the dc drive is enabled, the dc motor and the PMSM are in the stall condition. There is a constant error between the estimated and measured rotor positions. Once the dc drive is enabled, by using the information of the PMSM terminal voltages and phase currents, the estimated rotor position quickly converges to the measured value even in the low-speed range. After that the rotor position estimation error is always maintained within a constant range of ± 6 electric degrees same as that shown in Fig. 14, even when the torque increases from zero to the maximum value and then decreases back to zero. No torque-dependent offset is observed in the position estimation error. These results show that the proposed position estimator is robust to torque variations of the PMSM. Since the current commands are generated according to the MTPA profile, the amplitude of i_d has obvious changes during the torque transient, causing large spikes in φ with the amplitude around five electric degrees. This result is in coincidence with the simulation result shown in Fig. 4.

As shown in Fig. 13, the output torque of the PMSM first increases from zero to its maximum incrementally with a step of

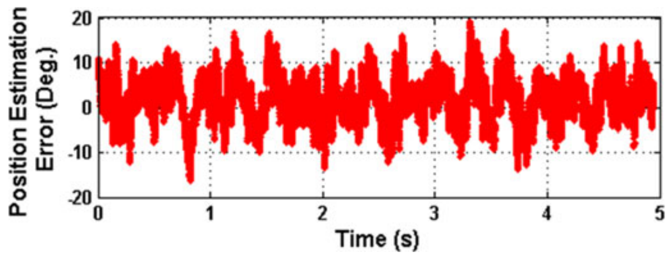

Fig. 12. Experimental results at 50 r/min and 20% load torque without the proposed dynamic position compensator. Error between the estimated (from the proposed estimator) and measured rotor positions.

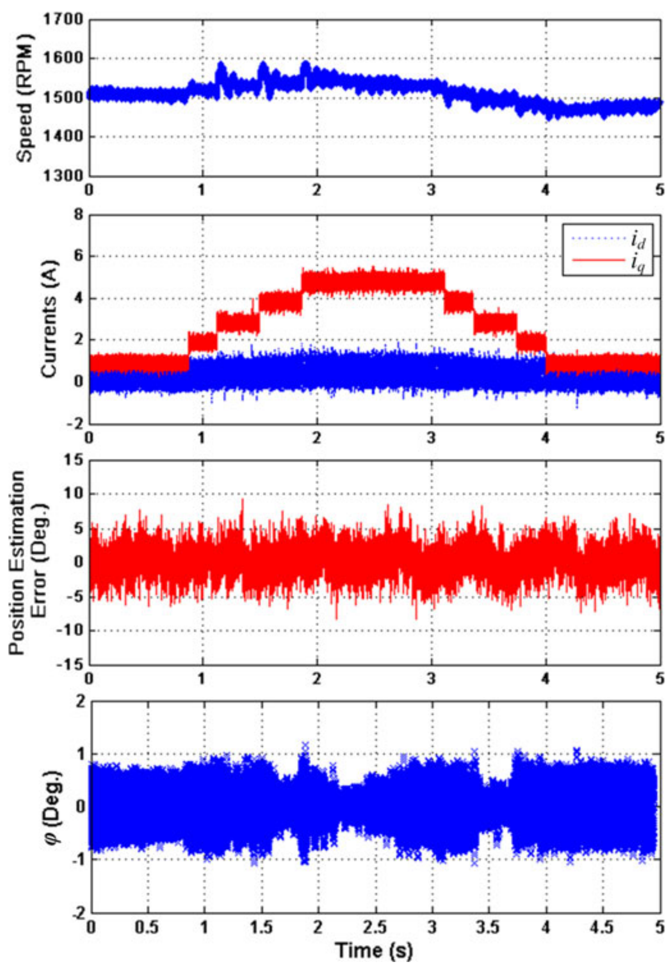

Fig. 13. Experimental results of the sensorless PMSM drive in the torque control mode using the proposed rotor position estimator by adjusting i_q only.

25% of the maximum output torque and then decreases to zero with the same steps. During the torque transition, the amplitude of i_q has step changes accordingly, while i_d stays at zero except for the ripples. The position estimation error is maintained within a constant range of ± 6 electric degrees and has no obvious dependence on the torque change. The amplitude of φ is always limited within ± 1 electric degrees during the test. It can be seen that the amplitude of φ has slightly change during the torque transit. This is caused by imperfect decoupling between

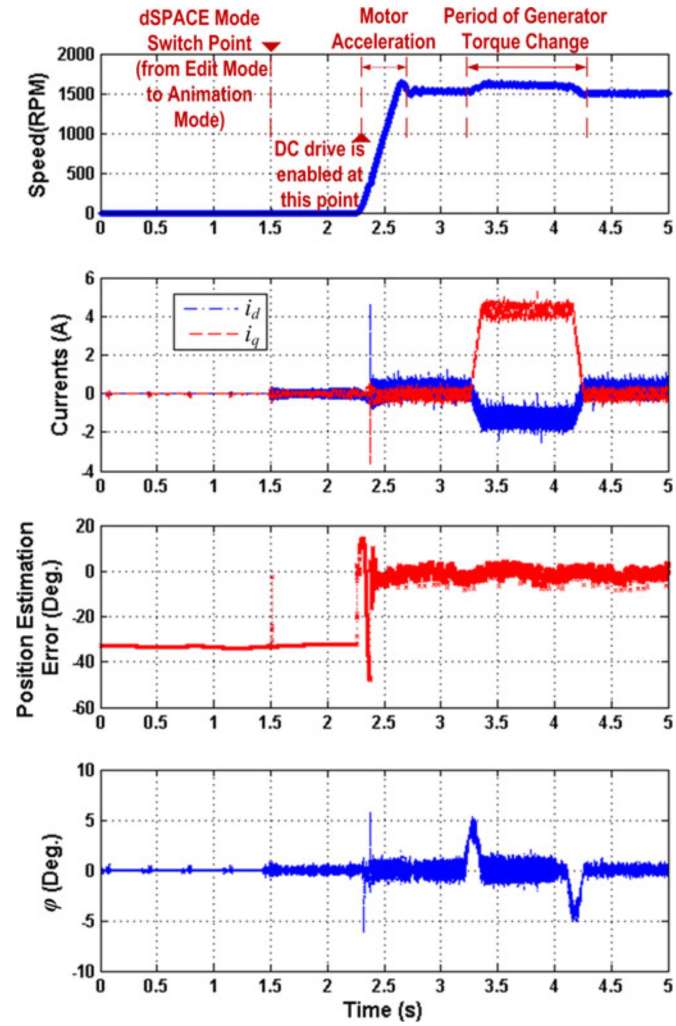

Fig. 14. Experimental results of the sensorless PMSM drive in the MTPA control mode using the proposed rotor position estimator.

the d- and q-control loops, such that the change in i_q slightly affects the magnitude of i_d.

VI. Conclusion

This paper has proposed a novel extended flux model-based rotor position estimator for sensorless control of salient-pole PMSMs. A comprehensive comparison has been provided to show that the proposed extended flux model has notable advantages of simpler structure and improved robustness to the variations of machine parameters and operating conditions (both speed and torque) when compared to the EEMF-based model. Extensive simulation and experimental results have been provided to validate the proposed rotor position estimator and sensorless control. Results have shown that, compared to the commonly used EEMF-based position estimators, the proposed estimator has much better dynamic performance, improved capability in very low-speed operating conditions, and is robust to speed and torque variations of the system.

References

[1] Z. Chen, M. Tomita, S. Doki, and S. Okuma, "An extended electromotive force model for sensorless control of interior permanent-magnet synchronous motors," *IEEE Trans. Ind. Electron.*, vol. 50, no. 3, pp. 288–295, Apr. 2003.

[2] S. Morimoto, K. Kawamoto, M. Sanada, and Y. Takeda, "Sensorless control strategy for salient-pole PMSM based on extended EMF in rotating reference frame," *IEEE Trans. Ind. Appl.*, vol. 38, no. 4, pp. 1054–1061, Jul./Aug. 2002.

[3] Y. Zhao, W. Qiao, and L. Wu, "An adaptive quasi-sliding-mode rotor position observer-based sensorless control for interior permanent magnet synchronous machines," *IEEE Trans. Power Electron.*, vol. 28, no. 12, pp. 5618–5629, Dec. 2013.

[4] S. Koonlaboon and S. Sangwongwanich, "Sensorless control of interior permanent-magnet synchronous motors based on a fictitious permanent magnet flux model," in *Proc. IEEE IAS Annu. Meet.*, Oct. 2005, pp. 311–318.

[5] I. Boldea, M. C. Paicu, and G. D. Andreescu, "Active flux concept for motion sensorless unified ac drives," *IEEE Trans. Power Electron.*, vol. 23, no. 5, pp. 2612–2618, Sep. 2008.

[6] J. Liu, T. A. Nondahl, P. Schmidt, S. Royak, and M. Harbaugh, "Rotor position estimation for synchronous machines based on equivalent EMF," *IEEE Trans. Ind. Appl.*, vol. 47, no. 3, pp. 1310–1318, May/Jun. 2011.

[7] G. Foo and M. F. Rahman, "Sensorless sliding-mode MTPA control of an IPM synchronous motor drive using a sliding-mode observer and HF signal injection," *IEEE Trans. Ind. Electron.*, vol. 57, no. 4, pp. 1270–1278, Apr. 2010.

[8] Y. Zhao, W. Qiao, and L. Wu, "Dead-time effect and current regulation quality analysis for a sliding-mode position observer-based sensorless IPMSM drives," in *Proc. IEEE IAS Annu. Meet.*, Oct. 2012, pp. 1–7.

[9] K. Liu and Z. Q. Zhu, "Online estimation of the rotor flux linkage and voltage-source inverter nonlinearity in permanent magnet synchronous machine drives," *IEEE Trans. Power Electron.*, vol. 29, no. 1, pp. 418–427, Jan. 2014.

[10] G. Wang, H. Zhan, G. Zhang, X. Gui, and D. Xu, "Adaptive compensation method of position estimation harmonic error for EMF-based observer in sensorless IPMSM drives," *IEEE Trans. Power Electron.*, vol. 29, no. 6, pp. 3055–3064, Jun. 2014.

[11] M. A. Ghazi Moghadam and F. Tahami, "Sensorless control of PMSMs with tolerance for delays and stator resistance uncertainties," *IEEE Trans. Power Electron.*, vol. 28, no. 3, pp. 1391–1399, Mar. 2013.

[12] D. Paulus, J. F. Stumper, and R. Kennel, "Sensorless control of synchronous machines based on direct speed and position estimation in polar stator-current coordinates," *IEEE Trans. Power Electron.*, vol. 28, no. 5, pp. 2503–2513, May 2013.

[13] Y. Zhao, W. Qiao, and L. Wu, "Sensorless control for IPMSMs based on a multilayer discrete-time sliding-mode observer," in *Proc. IEEE Energy Convers. Congr. Expo.*, Sep. 2012, pp. 1788–1795.

[14] Y. Zhao, W. Qiao, and L. Wu, "Improved rotor position and speed estimators for sensorless control of interior permanent magnet synchronous machines," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 2, no. 3, pp. 627–639, Sep. 2014.

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